

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 12****Topics covered :**

1. Linear and Homogeneous Differential Equations of order n
2. Fundamental and General Solutions

- Recall that:

1. The general form of an ordinary differential equation (ODE) of order n is given by $F(x, y, y', \dots, y^{(n)})$, and its normal form is written as $y^{(n)} = f(x, y, y', \dots, y^{(n-1)})$.
2. An ordinary differential equation is **linear** if it is a linear function of the unknown function y and its derivatives, i.e., $y, y', \dots, y^{(n)}$.
3. Any linear differential equation can be written as

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0(x)y = g(x), \quad (1)$$

where $a_i(x)$ for $i = 0, 1, \dots, n$ and $g(x)$ are functions of the independent variable x , but do not depend on the dependent variable y or its derivatives.

- The **differential operator** L (which can be considered a function involving differentiation) is defined as

$$L[y] = a_n(x)\frac{d^n y}{dx^n} + a_{n-1}(x)\frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x)\frac{dy}{dx} + a_0(x)y \quad (2)$$

Therefore, a linear ordinary differential equation (ODE) of order n can be written as $L[y] = g(x)$. It's important to note that L is a **linear operator**, as shown by the following

$$\begin{aligned} L[\alpha y_1 + \beta y_2] &= a_n(x)\frac{d^n (\alpha y_1 + \beta y_2)}{dx^n} + \dots + a_1(x)\frac{d (\alpha y_1 + \beta y_2)}{dx} + a_0(x) (\alpha y_1 + \beta y_2) \\ &= \alpha \left(\frac{d^n y_1}{dx^n} + \dots + a_1(x)\frac{dy_1}{dx} + a_0(x)y_1 \right) + \beta \left(\frac{d^n y_2}{dx^n} + \dots + a_1(x)\frac{dy_2}{dx} + a_0(x)y_2 \right) \\ &= \alpha L[y_1] + \beta L[y_2] \end{aligned} \quad (3)$$

- Let y_1 and y_2 denote two linearly independent solutions of the ODE $L[y] = g(x)$. By the property (3),

$$L[\alpha y_1 + \beta y_2] = \alpha L[y_1] + \beta L[y_2] = \alpha g(x) + \beta g(x) \quad (4)$$

Thus, $\alpha y_1 + \beta y_2$ is not necessarily a solution to $L[y] = g(x)$. However if $g(x) \equiv 0$ (i.e., 0 over the entire interval I) and y_1 and y_2 are linearly independent solutions to $L[y] = 0$, then

$$L[\alpha y_1 + \beta y_2] = \alpha L[y_1] + \beta L[y_2] = \alpha 0 + \beta 0 = 0 \quad (5)$$

Thus, $\alpha y_1 + \beta y_2$ is also a solution in this case.

- The linear differential equation $L[y] = g(x)$ is **homogeneous** if $g(x) \equiv 0$. From (5), we obtain the first result (or Theorem), which states that **for a linear homogeneous differential equation, if y_1 and y_2 are solutions, their linear combinations are also solutions.**
- A differential equation $L[y] = g(x)$ is called a **constant coefficient ODE** if the coefficients $a_n(x), \dots, a_0$ are constants. Note that $g(x)$ can still be a function of x .
- A collection of functions $y_1(x), \dots, y_n(x)$ are **Linearly dependent** on an interval I if there exists some constants $c_1, \dots, c_n$, not all zero, such that

$$\sum_{i=1}^n c_i y_i(x) \equiv 0 \quad (6)$$

over the entire interval. Functions that are not linearly dependent are called **linearly independent** functions. In the case of a collection of two functions, the condition to check for linear dependency is quite simple. Two functions y_1 and y_2 are linearly dependent if there exists a non-zero constant $\alpha \neq 0$ such that $y_1 = \alpha y_2$ over the entire interval.

- In general, checking if a given collection of functions are linearly dependent or not is quite challenging. However, if the given collection of functions are solutions to a linear ODE $L[y] = g(x)$ can be carried using the **Wronskian** of the collection of functions. The Wronskian of a collection of functions $y_1(x), \dots, y_n(x)$ is defined as

$$W(y_1(x), \dots, y_n(x)) = \begin{vmatrix} y_1 & y_2 & \cdots & y_{n-1} & y_n \\ y_1' & y_2' & \cdots & y_{n-1}' & y_n' \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_{n-1}^{(n-1)} & y_n^{(n-1)} \end{vmatrix} \quad (7)$$

For two functions y_1 and y_2 , the Wronskian is defined as

$$W(y_1(x), y_2(x)) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2 \quad (8)$$

- **Examples:**

1. The Wronskian of two functions [1] $y_1 = \cos(x)$ and $y_2 = \sin x$ is $W(y_1(x), y_2(x)) = \cos x \cos x - (-\sin x) \sin x = 1$.
2. The Wronskian of two functions [1] $y_1 = e^x$ and $y_2 = e^{2x}$ is $W(y_1(x), y_2(x)) = e^x (2e^{2x}) - e^x e^{2x} = e^{3x}$.

- The set of linearly independent solutions of an ODE are called the **Fundamental set of solutions**.

References

- [1] C. H. Edwards and D. E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th ed. Upper Saddle River, NJ: Pearson, 2000.